StyLitGAN: Enriching StyleGAN with Illumination Physics

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Figure 1. StyLitGAN searches for a set of directions in StyleGAN's $\mathbf{w}^+$ space that change generated image but not albedo then adds scaled versions of these directions to latents for other images to obtain relighted versions. There is no per-image search or finetuning. **Left:** images generated by a vanilla StyleGAN2. **Others** are relightings of the scenes in the leftmost image, with the same applied direction per column. Note extreme changes in the illumination of the scene; the geometry and albedo of the scenes do not change; and the images are realistic. Also note, each consistency in each column; directions have consistent effects across scenes.

Abstract

We enrich StyleGAN's generation capacity by relighting generated images. We use an intrinsic image method to decompose an image, then search the latent space of a pretrained StyleGAN to find novel directions that fix one component (say, albedo) and vary another (say, shading). Adding these directions to a latent vector results in a relighted image. Quantitative metrics of change in albedo and layout allow us to choose the intrinsic image method that produces the best results.

Our method, StyLitGAN, can add and remove luminaires in the scene and generate images with realistic lighting effects, including cast shadows, soft shadows, inter-reflections and glossy effects, without labeled, paired or CGI data. Qualitative evaluation confirms our results are realistic.

1. Introduction

The same scene viewed by the same camera will produce different images as the sun moves in the sky or people turn on or off lights. Similarly, changing the paint color on a wall will change the images of a scene. Current generative models can produce impressively realistic images from random vectors [15, 22–24], but cannot account for lighting effects. We show how to relight generated images to produce pictures of a given scene in a different light; and how to resurface scenes so that surface colors, but not layout or lighting, change. Our approach extends current editing methods [38, 45, 46, 52] by showing how to find directions that fix one intrinsic image component while changing another.

Our method is built on StyleGAN because there is an established procedure to manipulate style codes effectively to edit images. We use a pretrained and self-supervised

network to decompose an image into albedo, diffuse shading and glossy effects. We then search for edits to a style code that produce a set of images that (a) are diverse; and (b) have the same albedo (and so geometry and material) as the original generated image. Edits are achieved by adding constant vectors to the style code. Once directions with these properties have been found for some sets of images, we freeze the set and use it for all future images. Our edits can then be applied to style codes for any generated image. We find a form of consistency across images in our edits. So in Figure 1, adding the first direction tends to produce lighting from a directional source on the lower right (column Relighting-1); adding the third direction tends to produce very bright bedside lights (column Relighting-3); adding the fourth direction tends to produce dull non-directional lighting (column Relighting-4); and so on.

Our method can be seen as a GAN enrichment process. There really are many distinct lightings of the same scene, and these will not be present in the training data (and there is no evidence that a StyleGAN is willing to make multiple lightings of a scene unprovoked). StyleGAN's deep "knowledge" of image structure ensures that our images look realistic. Furthermore, many are unlike the training data because they show strong illumination effects. Similarly, by fixing lighting effects and changing albedo, we can get images that are unlikely to be present in training data because they show unusual color or color combinations. Our approach is easily adapted to produce a variant of StyleGAN – StyLitGAN - that produces a richer distribution of images by manipulating latent style codes and *which does not need to be trained*. We simply randomly choose an edit to apply to the style codes for the vanilla StyleGAN.

Performance is significantly affected by the choice of intrinsic image method. We choose an intrinsic image method by measuring the extent to which directions actually fix albedo and geometry and actually produce distinct images. The usual paraphernalia (FID, KID, etc.) of GAN evaluation does not apply in the usual way, because we are trying to generate realistic *out of distribution images*, so our FID *should* go up. We evaluate whether images are realistic qualitatively. Our method can generate any image that a vanilla StyleGAN can, by choosing not to edit. Further, we show that a vanilla StyleGAN cannot generate many of our images, and that StyLitGAN has a significant increase in FID over StyleGAN. This means the distribution of images generated by StyLitGAN has strictly larger support.

2. Related Work

Image manipulation: A significant literature deals with manipulating and editing images [3, 8, 9, 14, 16, 28, 33, 35, 44, 53]. Editing procedures for generative image models [15] are important, because they demand compact image representations with useful, disentangled, interpretations. StyleGAN [22–24] is currently de facto state-of-the-art for editing generated images, likely because it's mapping of initial noise vectors to style codes which control entire feature layers produces latent spaces that are heavily disentangled and so easy to manipulate. Recent editing methods include [7, 36, 38, 45, 46, 52], with a survey in [47]. The architecture can be adapted to incorporate spatial priors for authoring novel and edited images [11, 29, 43]. In contrast to this literature, we show how to fix one physically meaningful image factor while changing another. Doing so is difficult because the latent spaces are not perfectly disentangled, and we must produce a diverse set of changes in the other factor.

Relighting using StyleGAN: Stylerig [43] uses StyleGAN to relight faces but require 3D morphable face model. In contrast, StyLitGAN does not require any 3D model and extends to complex indoor scenes and Stylerig cannot. [48] uses semantic label attributes "indoor lighting" and "natural lighting" to train a binary classifier to find directions in latent space that represent them, but cannot produce diverse relighting. In contrast, StyLitGAN generates diverse, extreme, and realistic relighting without any labeled attributes. Other StyleGAN based face relighting methods – StyleFlow [1]; GAN control [39] – require that lighting can be expressed with a parametric model (in each case, spherical harmonics) but cannot relight rooms. In contrast, by forcing StyleGAN to produce room images with a fixed albedo (resp. shading field) we obtain images of relighted (resp. recolored) rooms that are (a) realistic and (b) it could not otherwise produce. We note that rooms are different and more challenging from faces: there are (a) significant long-scale inter-reflection effects; (b) shadow patterns are much more diverse and include soft shadows; (c) luminaires are often visible and are highly stylized; (d) surface albedos are stylized; and (e) surface brightnesses are not a function of normal alone. So environment maps or spherical harmonic models do not apply directly to rooms.

Other Face Relighting Methods mostly use carefully collected supervisory data from light-stages or parametric spherical harmonics [30, 32, 37, 41, 51]. ShadeGAN [31], Rendering with Style [5] and Volux-GAN [42] uses a volumetric rendering approach to learn the underlying 3D structure of the face and the illumination encoding. Volux-GAN [42] also requires image decomposition obtained from [32] that is trained using carefully curated light-stage data. In comparison, we neither require any explicit 3D modeling of the scene nor labeled and curated data for training image decomposition model [13].

Image Decomposition: Current image decomposition methods [4, 12, 13, 19, 27] decompose images into albedo and shading. Each score very well in current evaluations of WHDR (standard metric for intrinsic images due to [2]; recent summary of SOTA performance in [13]), but we have not found these decompositions to yield good relighting.

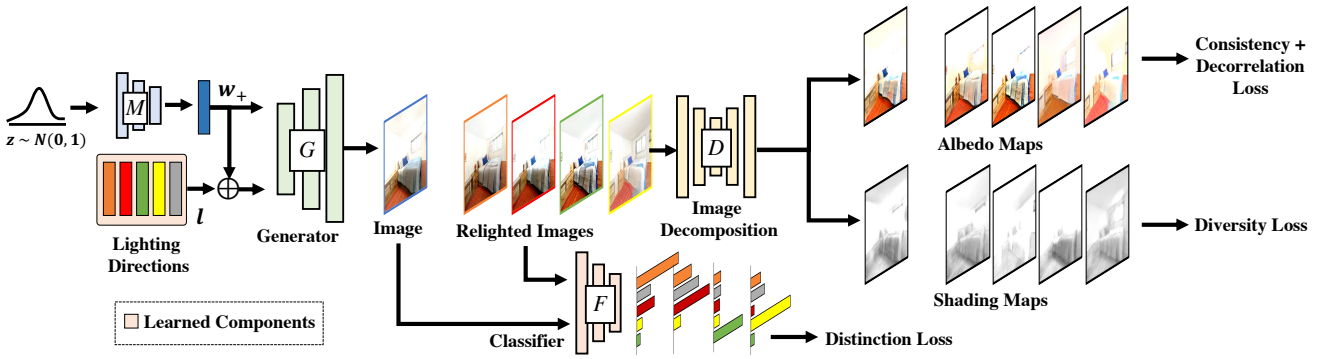


Figure 2. How StyLitGAN works: We generate an image from a random Gaussian noise using a pretrained StyleGAN. We also generate novel relighted images (16 in our case) of the same using randomly initialized latent directions ($\mathbf{d}$) that are added to $\mathbf{w}^+$ latent style codes. We train a classifier (F) that takes in all the pairs of relighted and original images and predicts the relighting direction applied to them. We apply a distinction loss and jointly update the latent directions and the classifier. Next, we generate the decomposition of these images from a pretrained decomposition model (D). We then apply losses that force StyLitGAN to find latent directions so that the albedo does not change (consistency loss), but the image does (diversity loss).

3. Approach

We follow convention and manipulate StyleGAN2 [24] by adjusting the $\mathbf{w}^+$ latent variables. We do not modify StyleGAN2, but instead, seek a set of lighting directions $\mathbf{d}_i$ which are independent of $\mathbf{w}^+$ and which have desired effects on the generated image. We obtain these directions by constructing losses that capture the desired outcomes, then search for directions that minimize these losses. We find all lighting directions at once and use 2000 randomly generated images to search for lighting directions. Our search procedure only sees each image once. Figure 2 summarizes our procedure and we call our model StyLitGAN.

Base StyleGAN Models: We use baseline pretrained models from [49] that use a dual-contrastive loss to train StyleGAN for bedrooms, faces and churches. We also use baseline pretrained StyleGAN2 models from [11] for conference rooms and kitchen, dining and living rooms.

Decomposition: We decompose images into albedo, shading and gloss maps (gloss; only when available) as $A \times S + G$, where A models albedo effects and S and G model shading and gloss effects respectively. We use the method of Forsyth and Rock [13], which is easily adapted because it uses only samples from statistical models that are derived from Land's Retinex theory [25] and is self-supervised. We can construct many decompositions using their approach by changing the statistical spatial model parameters. We evaluate several such decomposition models and choose the one that provides minimal albedo and geometry shift with a large relighting diversity (Section 4).

Relighting a scene should produce a new, realistic, image where the shading has changed but the albedo has not. Write $I(\mathbf{w}^+)$ for the image produced by StyleGAN given style codes $\mathbf{w}^+$, and $A(I)$, $S(I)$ and $G(I)$) for the albedo, shading and gloss respectively recovered from image I. We search for multiple directions $\mathbf{d}_i$ such that: (a) $A(I(\mathbf{w}^+ + \mathbf{d}_i))$ is very close to $A(I(\mathbf{w}^+))$ – so the image is a relighted version of $I(\mathbf{w}^+)$, a property we call **persistent consistency**; (b) the images produced by the different directions are linearly independent – **relighting diversity**; (c) which direction was used can be determined from the image, so that different directions have visibly distinct effects – **distinctive relighting**; and (d) the new shading field is not strongly correlated to the albedo – **independent relighting**. Not every shading field can be paired with a given albedo, otherwise, there would be nothing to do. We operate on the assumption that edited $\mathbf{w}^+$ will result in realistic images [6].

Recoloring: Alternatively, we may wish to edit scenes where the colors of materials of objects have changed, but the lighting hasn't. Because shading conveys a great deal of information about shape, we can find these edits using modified losses by seeking consistency in the shading field.

Persistent Consistency: The albedo decomposition of both the relighted scene: $A_R = A(I(\mathbf{w}^+ + \mathbf{d}_i))$ and the original: $A_O = A(I(\mathbf{w}^+))$ must be the same; where R refers to relighted images and O refers to StyleGAN generated images. We use a Huber loss and a perceptual feature loss [20, 50] from a VGG feature extractor (Φ) [40] at various feature layers (j) to preserve persistent effects (geometry, appearance and texture) in the scene.

$$\mathcal{L}_{const}(A_O, A_R) = \begin{cases} \frac{1}{2}\left[A_O - A_R\right]^2 & \text{for } |A_O - A_R| \leq \delta, \\ \delta\left(|A_O - A_R| - \delta/2\right) & \text{otherwise.} \end{cases} \quad (1)$$

$$\mathcal{L}_{per}(A_O, A_R) = ||\Phi_j(A_O) - \Phi_j(A_R)||_2. \quad (2)$$

Relighting Diversity: We want the set of relighted images produced by the directions to be diverse on a long scale so that regions that were in shadow in one image might be bright in another. For each $S(\mathbf{w}^+ + \mathbf{d}_i)$, we stack the two shading and gloss: S and G and compute a smoothed and

downsampled vector $\mathbf{t}_i$ from these maps. We then compute $\mathcal{L}_{div}(S,G)$ (diversity loss) which compels these $\mathbf{t}_i$ to be linearly independent and encourages diversity in relighting.

$$\mathcal{L}_{div}(S,G) = -\log\det N \tag{3}$$

where i^{th} & j^{th} component of N is $t_i^\intercal t_j$

Distinctive Relighting: A network might try to cheat by making minimal changes to the image. Directions $\mathbf{d}_i$ should have the property that $\mathbf{d}_i$ is easy to impute from $I(\mathbf{w}^+ + \mathbf{d}_i)$. We train a classifier joint with the search for directions. This classifier accepts $I(\mathbf{w}^+)$ and $I(\mathbf{w}^+ + \mathbf{d}_i)$ and must predict i. The cross-entropy of this classifier supplies our loss:

$$\begin{aligned}&\min_{l,F} \mathcal{L}_{dist}(I(\mathbf{w}^+),\ I(\mathbf{w}^+ + d_i))\\ &= -\sum_{i=1}^{M} d_i \log F(I(\mathbf{w}^+),\ I(\mathbf{w}^+ + d_i))\end{aligned} \tag{4}$$

Independent Relighting: Intrinsic image methods tend to report somewhat correlated albedo and shading (dark image patches tend to be interpreted as both dark albedo and dark shading). No current method reliably avoids this class of error. While the error does not seem to have a major effect on WHDR scores, it significantly weakens relighting diversity, because lighter objects will tend to be brighter, and darker objects will tend to be darker. We discourage this effect by decorrelating the surface lightness (measured at a long scale) from relighted images. We assume surface lightness can be measured from albedo (average of RGB channels) and add a small penalty if it is like the intensity of relighted images, using the negative of cosine similarity on the smoothed version of the gray-scale albedo and relighted images. This allows the albedo to deviate slightly from the original image and still be consistent with the original scene without making large color shifts.

$$\mathcal{L}_{deco} = -\frac{A_O \cdot I(\mathbf{w}^+ + d_i)}{|A_O||I(\mathbf{w}^+ + d)|} \tag{5}$$

The overall loss is

$$\lambda_{const}\mathcal{L}_{const} + \lambda_{per}\mathcal{L}_{per} + \lambda_{div}\mathcal{L}_{div} + \lambda_{dist}\mathcal{L}_{dist} + \lambda_{deco}\mathcal{L}_{deco} \tag{6}$$

Recoloring requires swapping persistent for transient components in all losses, except we do not use decorrelation loss while recoloring. Obtaining good results requires quite a careful choice of loss weights (λ coefficients). Corresponding λ coefficients for both these edits are in our supplementary.

4. Choosing an Intrinsic Image Decomposition

We choose an intrinsic image decomposition that yields directions that: (a) do not change albedo; (b) do not change geometry; and (c) strongly change the image. We must choose qualitatively based on a plot because there is no way to weigh these effects against one another. However, relatively few methods are admissible.

Scoring albedo change: We use SuperPoints [18] to find 100 interest points in the original StyleGAN generated image. Around each interest point, we form a 10×10 patch. We then compare these patches with patches in the same locations for multiple different relightings of that image. If the albedo in the image does not change, then each patch will simply be relighted. Now consider the patch $p^o(x,y)$ at (u,v) in the original image, and compare to the corresponding patch in the i'th relighted image of that scene $p^i(x,y)$. If the albedos are the same ($\rho(x,y)$), we expect that $p^o(x,y) = \rho(x,y)s^o(x,y)$ and $p^i(x,y) = \rho(x,y)s^i(x,y)$, for $s(x,y)$ shading fields. Further, because the patches are small, we expect the shading fields are spatially simple. In particular, we model $s(x,y)$ as a linear ramp, so $s^i(x,y) = a_i + b_i x + c_i y$. Using this model, if the two patches have the same albedo, then there should be $a_o \ldots c_i$ such that $p^o(x,y)(a_i + b_i x + c_i y) - p^i(x,y)(a_o + b_o x + c_o y) = 0$. Note this is homogeneous. We measure albedo shift by

$$\min_{a_o,\ldots,c_i} \sum_{x,y\in\text{patch}} \left[\begin{array}{l} p^o(x,y)(a_i + b_i x + c_i y) - \\ p^i(x,y)(a_o + b_o x + c_o y)\end{array}\right]^2$$

subject to $a_o^2 + \ldots c_i^2 = 1$. This is an eigenvalue problem.

Scoring Geometry Change: We score geometry changes by scoring the change in the surface normal for the relighted image. This presents a technical problem. Normals predicted by even very strong current methods change when lighting changes [21]. Rather than resolve this difficulty, we used StyleGAN itself to predict normals. Write $G(\mathbf{w})$ for the image generated by $\mathbf{w}$. We search for a direction $\mathbf{d}$ so that $G(\mathbf{w}+\mathbf{d})$ is the normal map of $G(\mathbf{w})$. This direction is fixed and is not image-dependent. The direction is chosen so that $G(\mathbf{w}+\mathbf{d})$ is consistent with Omnidata's predicted normal for $G(\mathbf{w})$ [10]. Surprisingly, such a direction can be found relatively easily, and finetuning or modifying the generator is unnecessary. This normal prediction is relatively robust to change in illumination compared to Omnidata's normals. We then score geometry shifts by directly evaluating the variance in surface normals for all relighted images. High variance indicates a large geometry shift, and low variance indicates a small geometry shift.

Scoring Lighting Diversity: For diversity in lighting we directly estimate the average mean-squared error difference between our relighted images and the original StyleGAN generated images.

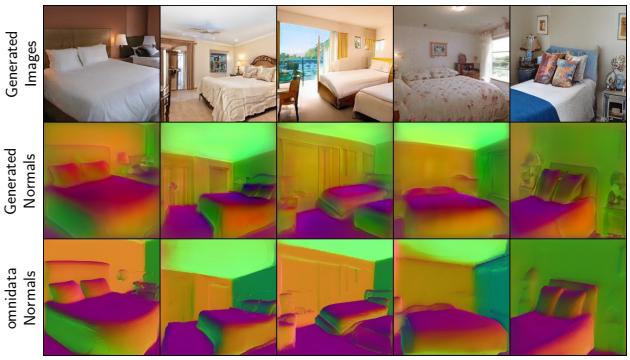


Figure 3. Normals can be *generated* directly from StyleGAN by searching *only* as style directions. No finetuning or architecture modification is required. The top row is StyleGAN generated images, the middle row is normals generated from the StyleGAN and the bottom row are normals from Omnidata [10]. We use [10]'s normals to guide our search. Note accurate and comparable generated normals to Omnidata. We find a similar direction for depth (supplementary).

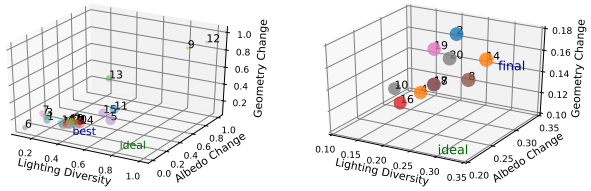
Decomposition Models Investigated: We searched 20 instances in total obtained with different hyperparameter settings from three families of decomposition. The first family is the SOTA unsupervised model of [13], which decomposes images into albedo and shading using example images drawn from statistical models. The second is a variant of that family that decomposes into albedo, shading and gloss decomposition. The third is an albedo, shading and gloss decomposition that models fine edges in the albedo rather than the shading field. These models were chosen to represent a range of possible decompositions, but others could yield better results. The key point is that we can choose a model from a collection by a rational process.

Selecting a Model: We create a 3D scatter plot between lighting diversity, albedo change and geometry change and choose the model that has large lighting diversity with moderate geometry and albedo change (Figure 4). Note from the figure that some models are clearly inadmissible (behind the Pareto frontier).

Selecting Directions: For a given model, some directions may yield larger (or smaller) changes in albedo or geometry. We use the procedure of Section 3 to find 16 base directions, then choose the 10 of those that produce the smallest albedo and geometry changes.

5. Experiments

Qualitative evaluation: The primary goal of StyLitGAN is to produce realistic images, that are out of distribution but known to exist for straightforward physical reasons. Because they're out of distribution, current quantitative evaluation



All Models Zoomed in

Figure 4. We compare different decomposition, hyperparameter choices, and select models that yield large relighting diversity with small albedo and geometry change. The change in albedo is measured by the variation in small local patches, the change in geometry is measured by the variation in surface normals and the change in lighting is measured by the shift in pixel intensity. Only a few choices are admissible. Admissible models are on the right. We finalize Model 14 and all other figures are from that model.

Table 1. FID measures distribution shift and not realism. Our generated images are realistic and are out-of-distribution because of large illumination and color changes in the images. This results in large FID scores. KDL in the table is for kitchen, dining and living room which are jointly trained [11].

Type	Bedroom	KDL	Conference	Church	Faces
StyleGAN (SG)	5.01	5.86	9.35	3.80	5.02
SG + Relighting (RL)	14.23	6.87	10.48	12.12	37.87
SG + Resurfacing (RS	17.03	9.41	10.63	18.60	34.06
SG + RL + RS	21.39	11.68	12.71	21.08	37.40

tools do not apply. We evaluate realism qualitatively. Further, there is no direct comparable method. However, we show relighting comparisons to a recent SOTA method that is physically motivated and trained with CGI data [26] (Figure 10). For relighting, our method should generate images that: are clearly relightings of a scene; fix geometry and albedo but visibly change shading; and display complicated illumination effects, including soft shadows, cast shadows and gloss. For resurfacing, our method should generate images that: are clearly images of the original layout, but with different materials or colors or changes in furniture; and display illumination effects that are consistent with these color changes. As Figure 6 shows, our method meets these goals. Figure 7 shows we can relight scenes with a novel GAN inversion method. Figure 8 and Figure 9 show interpolation sequences for a relighting (resp. recoloring). Note that the lighting (resp. coloring) changes smoothly, as one would expect. Figure 11 shows our relighting and recoloring directions are largely disentangled.

Generality: We have applied our method to StyleGAN instances trained on a variety of different datasets – Conference Room, Kitchen, Living Room, Dining Room and Church and Faces (results in supplementary; relighting is successful for each).

original d_1 d_2 d_3 d_4 d_5 d_6 d_7 d_8 d_9 d_{10}

Figure 5. **First columns** images generated by the original StyleGAN. **Other columns** show images obtained from $\mathbf{w}^+ + \mathbf{d}_i$ for our top 10 relighting directions for the chosen model added to the style codes ($\mathbf{w}^+$) corresponding to the the image on the first column's style codes ($\mathbf{w}^+$). These directions have been chosen to fix albedo, but change transient illumination properties – shading and gloss. Note: each column appears to show the same scene that is the row with different illumination; lighting varies aggressively, and the individual latent direction $\mathbf{d}_i$ have persistent semantics – each column corresponds to a type of illumination.

Figure 6. **First columns** images generated by the original StyleGAN. **Other columns** show images obtained from $\mathbf{w}^+ + \mathbf{d}_i$ for our top 10 resurfacing or recoloring directions for the chosen model added to the style codes ($\mathbf{w}^+$) corresponding to the the image on the first column. These directions have been chosen to fix transient effects (overall illumination) and scene layout but change albedo (surface color or materials). Note: each column appears to show the same scene that is in the row with the same illumination; color or surfaces varies aggressively, and the individual latent direction $\mathbf{d}_i$ have persistent semantics – each column corresponds to a type of surfacing choice.

Figure 7. We can relight real scenes using a novel GAN inversion method (supplementary). Note persistent lighting semantics are preserved after inversion and relighting.

Figure 8. Interpolation in relighting directions. Top row: linear scaling from -1 to 1 of a random direction and bottom row interpolation between two random relighting directions. Linear scaling along a single direction results in a complicated illumination change, from near uniform to pointed direction light moving left to right in the top row. However, interpolation between two directions appears to follow a path – light moving anti-clockwise with a strong gloss on the bottom left corner of the floor to the bottom right corner of the bed sheet.

Quantitative evaluation: Figure 4 shows how we can evaluate albedo shift, geometry shift and lighting shift. Further, we show we can generate image datasets with *increased* FID [17, 34] compared to the base comparison set in Table 1 (we use clean-FID [34]). This is strong evidence our method can produce a set of images that is a strict superset of those that the vanilla StyleGAN can produce.

6. Discussion

StyleGAN's relighting suggests it "knows" quite a lot about images. It knows where bedside lights and windows

Figure 9. Interpolation in recoloring directions. Top row: linear scaling from -1 to 1 of a random direction and bottom row interpolation between two random recoloring directions. Linear scaling using a single direction appears to be constructing and deconstructing the scene. While interpolating between two directions results in smooth albedo changes in the scene and expansion of the bed.

Figure 10. With an appropriate inversion method (supplementary), we can relight real scenes. **Top row:** shows three images of the same scene in different lightings, obtained from [26]. **Middle row:** shows relightings of the first image, obtained by Li *et al.* by inverse rendering, changing luminaire parameters, then forward rendering [26]. **Bottom left:** image generated by passing latent variables from our inversion of the top left through StyleGAN. **Bottom center** and **bottom right** show relightings obtained by adding a relighting direction to these latent variables. Recall the relighting directions are *independent of image*. The directions were selected by hand to correspond to the top center, right respectively; directions are Relighting - 1 (visible light off) and Relighting - 3 (invisible light off) from Figure 1. Note: our relights compare well to real images; our relights do not ring on fine edges (eg the lamp); our relights preserve high spatial frequencies in the image; and our relights are innocent of CGI or physical rendering.

are, and can be persuaded to produce normal maps. It is intriguing to wonder how well complex scene properties could be bullied out of an image generator. Is StyleGAN and an inverter all we need to recover maps of scene properties?

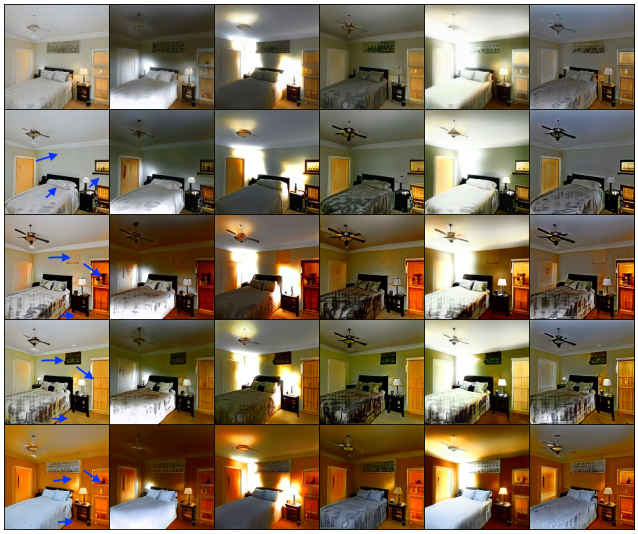


Figure 11. We show what happens when we use both relighting and resurfacing directions for a single scene in StyLitGAN. **Columns** show different relights for a fixed resurfacing and **rows** show different resurfacing for a fixed relight. There is clearly some interaction, but not much - they are largely disentangled. Lighting and large color changes are more dramatic and easy to visualize. Small surface changes are highlighted in blue. (Best when viewed on screen)

Choice of decomposition model matters: Our experiments suggest studying intrinsic image decomposition from a utilitarian perspective – the best method is one that helps solve a problem.

Targeted Relighting: Our method generates random and diverse relighting, but cannot produce targeted relighting. Another important limitation is that, although we have demonstrated our method can produce images that StyleGAN can't but should be able to, we do not know how to show that we have *all* such images. We found new images by imposing a simple, necessary property of the distribution of images (scenes have many different lightings) and we expect that other such observations could lead to other kinds of good but out-of-distribution images.

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